Geological Sequestration: Basic Principles of Financial Assurance

In 2010, the United States Environmental Protection Agency (USEPA) published Federal requirements for Class VI Geologic Sequestration wells at 40 CFR 146, Subpart H.

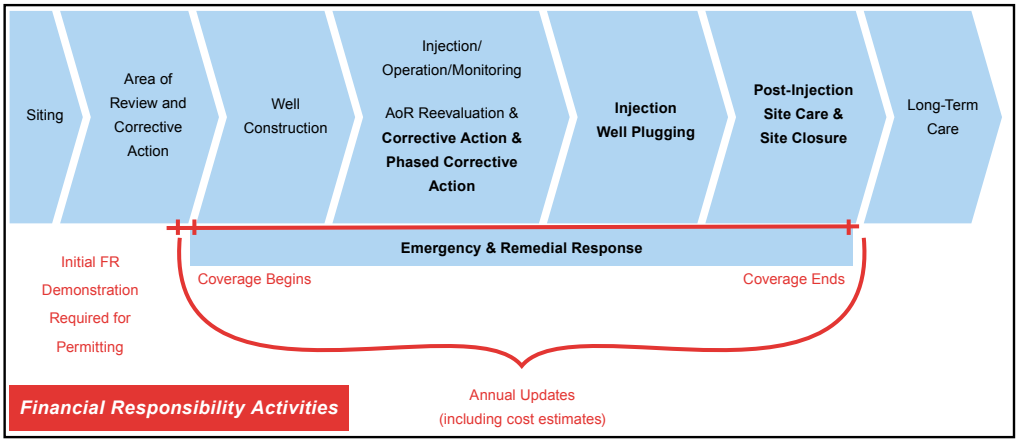


Figure 1. UIC Class VI Program, Covered Activities. Source: EPA 816-R-11-005

As a companion document to the regulations, the USEPA issued *Underground Injection Control (UIC) Program Class VI Financial Responsibility Guidance* in July 2011. Through these efforts, the USEPA addresses site characterization, well construction, monitoring of well integrity, injection, storage, ground water quality, and post-injection site care.

Collectively, the two-tiered framework establishes a paradigm, under which owners and operators of Class VI geologic sequestration wells must demonstrate the financial means to cover the costs of: (1) well construction; (2) injection, operations & monitoring (e.g., Area of Review reevaluation & corrective action), (3) injection well plugging, (4) post-injection site care & site closure, and (5) emergency & remedial response.

An effective financial assurance framework will assure funds are available to pay for activities necessary to detect problems before they adversely impact public welfare or the environment, and finance remedial action should such action become necessary. The USEPA's financial responsibility guidance offers suggested standards for financial responsibility, as well as recommendations related to the wording of key financial terms, procedures for funding financial instruments, and procedures for drawing down (or reimbursing costs) from financial instruments.

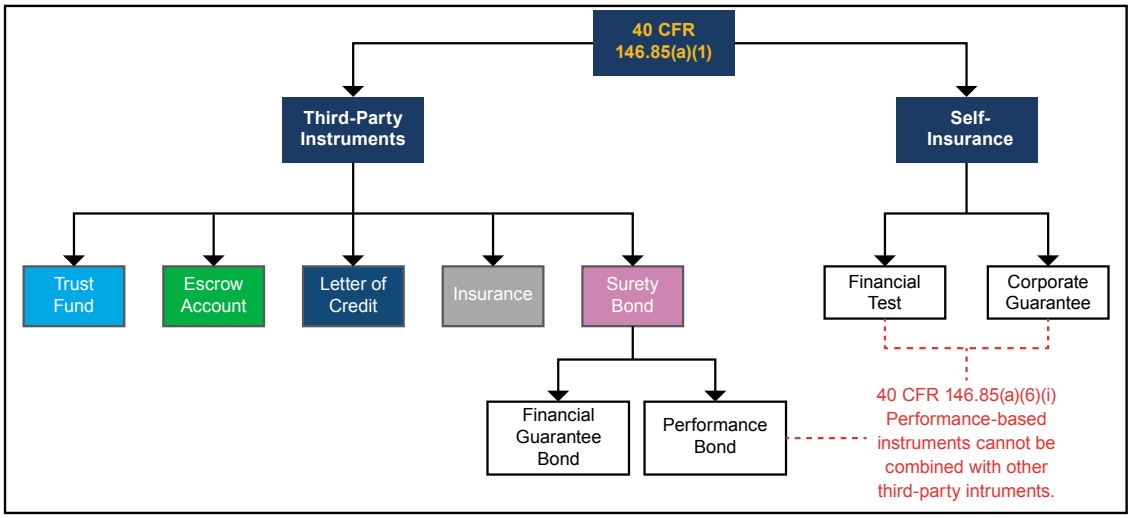
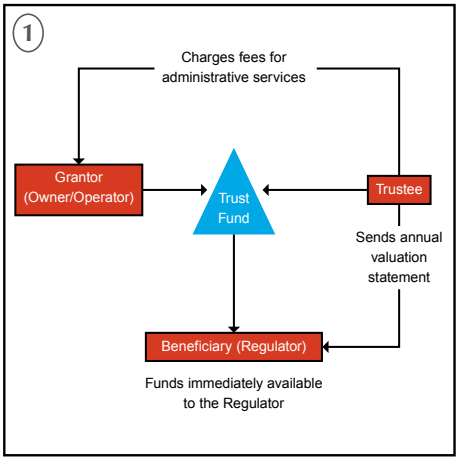
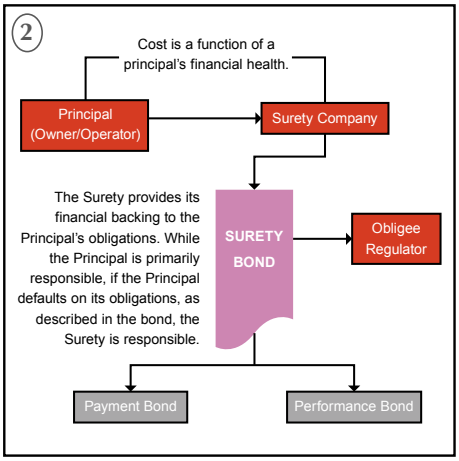
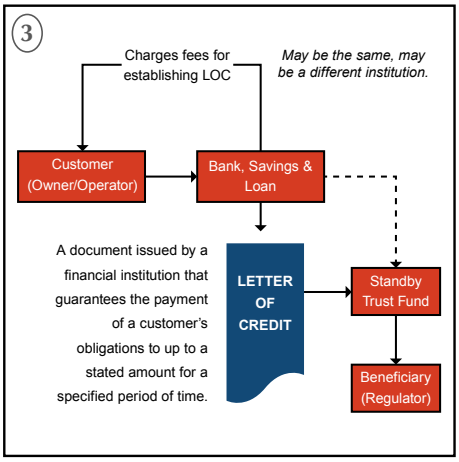


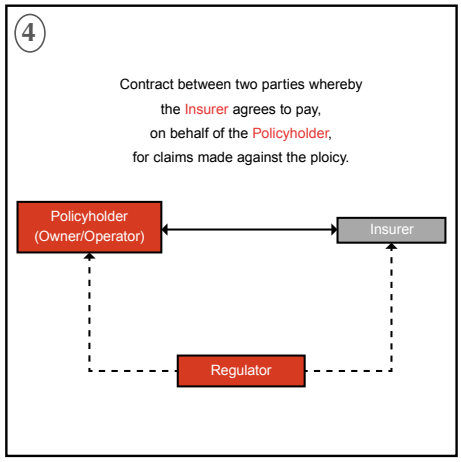
Figure 2. Qualifying Class VI Financial Instruments. See next page for more detailed descriptions of Financial Instruments. Source: STEP, FA Workshop

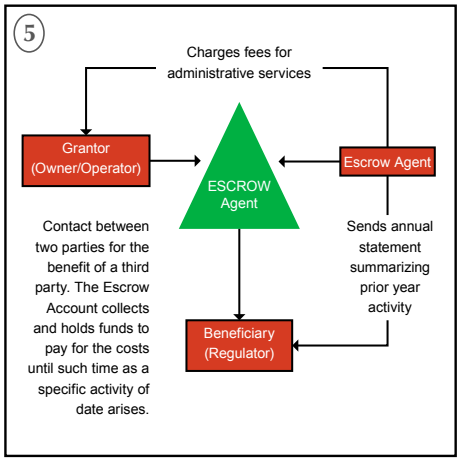
Specifically, the USEPA guidance recommends that UIC Class VI well owners and operators demonstrate financial responsibility through one of six financial instruments (shown below), including: (1) Trust Fund; (2) Surety Bond; (3) Letter of Credit; (4) Insurance Policy; (5) Escrow Account; and (6) Self-Insurance (e.g., Financial Test or Corporate Guarantee). Across the third-party instruments, well operators also can demonstrate financial assurance through a hybrid of financial instruments. Financial assurance stimulates firms to appropriately internalize costs, and in so doing, prevent excessive consumption of environmentally damaging products.











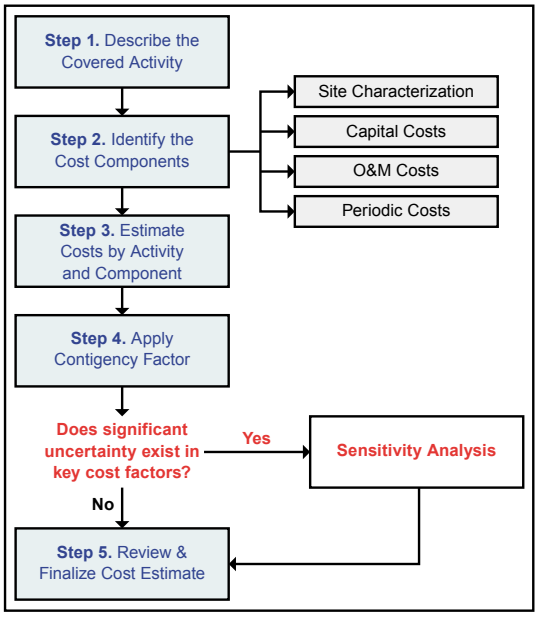


Figure 3. Cost Estimation

The market impacts associated with financial responsibility range from enhanced capital investment, such that firms design site projects to minimize adverse environmental costs, to deterrence, and such that firms make operating decisions that consider how best to reduce the likelihood of environmental injury. The face value of financial assurance is informed by accurate and current cost estimates. In general costs, and therefore financial assurance, are a function of plant design and location, and should be estimated on a site specific basis.

The choice of which financial instrument will be most effective depends on the near- and long-term goals of the project. For example, when considering financial assurance for long-term care, the objective should be to ensure sufficient funds exist to pay for care, including operating and maintenance costs on a going forward basis. In many instances, Trust Funds, or dedicated interest-bearing accounts, whereby the instrument is funded on a graduated basis over a defined period of time, tend to be best suited to provide financial assurance for activities occurring further out in time and/or over a long period of time. Conversely, self-insurance instruments, like the Corporate Financial Test, tend to be best suited assure activities occurring in the near to medium term, i.e., during the operating life of the project.